Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

(a) Interpretation of the Scenario and Type of Noise

The description of "repetitive horizontal stripes" caused by electrical interference is the classic signature of periodic noise. Periodic noise is introduced when the imaging or power system picks up interference from electrical or electromechanical sources — for example, AC power line hum, switching power supplies, motor vibrations in the conveyor mechanism, or poorly shielded camera electronics. Unlike random noise (such as Gaussian or salt-and-pepper noise), periodic noise is structured: it repeats at a fixed spatial frequency and orientation across the entire image, which is exactly why it appears as regular horizontal bands rather than isolated random spots.

Because the interference is coupled into the sensor's row-scanning or power circuitry, it manifests as a sinusoidal-like intensity variation running across rows (horizontal stripes), repeating at a constant interval down the image. This is the key diagnostic clue that separates periodic noise from other noise categories:

- Repetitive pattern → rules out purely random noise (Gaussian, Poisson, impulse).
- Fixed orientation and spacing → indicates a dominant, narrow-band frequency component.
- Electrical origin → consistent with power-line or clock-signal coupling into the imaging hardware.

(b) Effect of the Degradation on Inspection Accuracy

Automated defect detection algorithms typically rely on local intensity gradients, texture analysis, or edge detection to identify scratches, cracks, dents, or surface irregularities on metal components. Periodic stripe noise directly interferes with these mechanisms in several ways:

1. False positives: the repeating stripe pattern creates artificial edges and intensity transitions that can be mistaken for real surface defects, increasing the false-alarm rate and causing good components to be rejected.
2. False negatives: genuine low-contrast defects (fine scratches, hairline cracks) can be masked or blended into the stripe pattern, especially where a defect's orientation coincides with the stripe direction, causing real defects to be missed.
3. Reduced measurement reliability: any dimensional or texture-based measurement (surface roughness, edge-position measurement) becomes biased because the periodic component adds a systematic, non-random error across every image.
4. Inconsistent thresholding: intensity-based thresholding techniques, common in inspection pipelines, become unreliable because the effective background intensity now varies periodically rather than remaining flat.

Overall, the interference does not just reduce visual quality — it corrupts the very features (edges, gradients, local contrast) that the inspection algorithm depends on, making the system unreliable and unsuitable for pass/fail decisions on a production line.

(c) Recommended Enhancement Technique

The most appropriate technique is frequency-domain filtering using the Fourier Transform, specifically a Notch Reject Filter (or Band-Reject Filter for a broader periodic band). The procedure is as follows:

5. Transform the corrupted image from the spatial domain to the frequency domain using the 2-D Discrete Fourier Transform (DFT / FFT).
6. Inspect the magnitude spectrum. Periodic noise appears as isolated, symmetric bright spots (impulses) at specific frequency locations away from the origin, corresponding to the frequency and orientation of the stripe pattern.
7. Design a notch reject filter (ideal, Butterworth, or Gaussian notch) that attenuates only the narrow frequency band at those noise-spike locations, while passing all other frequencies unaffected.
8. Multiply the filter with the frequency-domain representation of the image to suppress the noise spikes.
9. Apply the Inverse Fourier Transform (IDFT) to reconstruct the cleaned image back in the spatial domain.

A Gaussian or Butterworth notch filter is generally preferred over an ideal notch filter because it avoids ringing artefacts (introduced by the sharp cut-off of an ideal filter), giving a smoother, more visually and analytically accurate restoration.

(d) Justification: Frequency Domain vs. Spatial Domain

Spatial-domain filters (mean, median, Gaussian smoothing) operate uniformly across the image and are effective against random noise, but they are poorly suited to periodic noise for several reasons:

Criterion	Spatial-Domain Filtering	Frequency-Domain Notch Filtering
Noise localisation	Cannot isolate a specific repeating pattern; treats noise as generic local variation	Periodic noise appears as sharp, isolated spikes that can be precisely located and targeted
Edge / detail preservation	Averaging (mean/median) blurs fine defect edges while only partially reducing striping	Only the narrow noise frequency is removed; all other frequencies, including fine defect edges, remain intact
Effectiveness on structured noise	Weak — stripes are broadband in orientation and survive local smoothing	Strong — the DFT concentrates periodic energy into a few identifiable frequency components that can be surgically removed
Risk of side effects	Requires aggressive smoothing to suppress stripes, degrading defect visibility	Minimal, since filtering is confined to the exact noise frequencies

In short, periodic noise is a frequency-selective problem, so it requires a frequency-selective solution. Spatial filtering would need to blur the whole image heavily to suppress the stripes, destroying the very edge and texture information the inspection system needs to detect defects. Frequency-domain notch filtering, by contrast, surgically removes only the noise-causing frequency components and leaves the rest of the image detail — including subtle defect edges — untouched.

(e) Logical Sequence of the Solution

10. Recognise the stripe pattern as periodic noise arising from electrical interference.
11. Understand that periodic noise corrupts edge and gradient information, directly harming defect-detection accuracy.
12. Conclude that a global, non-selective spatial filter would blur real defects while only partly reducing the stripes, and is therefore inadequate.
13. Transform the image to the frequency domain to reveal the noise as isolated spectral spikes.
14. Apply a notch/band-reject filter to eliminate exactly those spikes.
15. Inverse-transform back to the spatial domain, producing a stripe-free image with defect edges preserved, ready for reliable automated inspection.

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

(a) Interpretation of the Scenario and the Challenge Involved

Heavy rain degrades images captured by a vehicle's cameras in multiple, overlapping ways: raindrops on the lens or windshield scatter and refract light, water spray from the road creates fine random speckling, and low ambient light combined with high sensor gain introduces additional random sensor noise (commonly modelled as Gaussian noise, with an impulsive/speckle-like component from individual droplets and spray). Unlike the previous scenario, this noise is largely random and irregular rather than periodic.

The central challenge is that noise removal cannot be pursued in isolation — the corrected image must still allow the perception system to reliably detect lane markings and road edges, since the vehicle's steering and path-planning decisions depend directly on those features. This creates a dual, and partly conflicting, requirement: smooth out the random rain-induced noise, but do not blur or displace the sharp intensity transitions that define lane boundaries.

(b) Importance of Preserving Edges While Removing Noise

Lane markings and road edges are detected primarily through edge and gradient information — the sharp intensity transition between the road surface and a painted lane line, or between the road and its shoulder. Standard noise-suppression filters (e.g., simple averaging/mean filters or large-kernel Gaussian blur) reduce noise by averaging neighbouring pixels, but this same averaging also smooths out sharp edges, effectively blurring lane boundaries.

In an autonomous vehicle, this has direct safety consequences:

- Blurred or weakened lane edges reduce the confidence and accuracy of lane-detection algorithms, increasing the risk of lane-departure or incorrect path-following.
- Softened road-edge boundaries make it harder to distinguish drivable surface from shoulder or obstacles, especially in low-visibility rain conditions.
- Delayed or inaccurate edge information directly affects real-time control decisions (steering angle, braking), where even small localisation errors can compound into unsafe manoeuvres.

Therefore, an enhancement approach for this scenario must be edge-preserving by design — it must distinguish between noise (which should be smoothed) and true structural edges (which must remain sharp).

(c) Recommended Filtering Technique

The most suitable technique is the Bilateral Filter (an edge-preserving smoothing filter). As an alternative or complement, median filtering can be used specifically for the impulsive/speckle component from droplets, but the bilateral filter is the primary recommendation because it directly balances noise removal against edge preservation. The bilateral filter works as follows:

16. For each pixel, a weighted average of neighbouring pixels is computed, similar to Gaussian smoothing.
17. However, two weighting functions are combined: a spatial (domain) kernel based on geometric distance from the centre pixel, and a range (intensity) kernel based on how similar the neighbour's intensity is to the centre pixel's intensity.
18. Pixels that are spatially close AND have similar intensity (i.e., lie on the same smooth surface) are averaged together, suppressing random noise.
19. Pixels that are spatially close but have very different intensity (i.e., lie across an edge, such as a lane line boundary) receive a very low weight, so the edge is not averaged away.
20. The result is a smoothed image where random rain-induced noise is suppressed while the sharp lane-marking and road-edge transitions remain intact.

Where the noise includes a strong impulsive component (isolated bright/dark specks from droplets or spray), a median filter can be applied as a pre-processing or complementary step, since it removes such outliers without significantly blurring edges either — though the bilateral filter remains preferable for the overall Gaussian-like rain noise while preserving fine gradients.

(d) Justification: Why This Filter Preserves Important Detail

Criterion	Simple Averaging / Gaussian Blur	Bilateral Filter
Weighting basis	Spatial distance only	Spatial distance AND intensity similarity
Behaviour at edges	Averages across edges, causing blur	Suppresses averaging across large intensity jumps, keeping edges sharp
Noise reduction in flat regions	Effective	Equally effective, since nearby pixels are also intensity-similar in flat regions
Suitability for lane detection	Weakens lane-line boundaries, risking detection errors	Preserves lane and road-edge boundaries, supporting reliable detection

The core justification is that the bilateral filter is content-adaptive: its behaviour changes locally depending on whether the neighbourhood is homogeneous (likely noise, safe to smooth) or contains a strong intensity transition (likely a real edge, must be preserved). This adaptive behaviour is precisely what a linear spatial filter (mean or Gaussian) cannot provide, since a linear filter applies the same weighting rule everywhere regardless of local image content. For a safety-critical application such as autonomous navigation, this trade-off — effective denoising without sacrificing the edges the system relies on — makes the bilateral filter (optionally combined with median filtering for impulsive droplet noise) the more suitable choice over both plain smoothing filters and non-edge-aware frequency-domain methods.

(e) Systematic Explanation

21. Heavy rain introduces a mix of random noise (droplet scatter, spray, low-light sensor noise) onto road images.
22. The vehicle's safety depends on lane markings and road edges — features defined by sharp intensity gradients — remaining clearly detectable.
23. A naive denoising filter (mean/Gaussian blur) would reduce noise but also blur these critical edges, which is unacceptable for navigation.
24. An edge-preserving filter is therefore required: one that adapts its smoothing based on local intensity similarity, not just spatial proximity.
25. The bilateral filter satisfies this requirement by combining a spatial kernel with a range (intensity) kernel, smoothing flat/noisy regions while leaving strong edges untouched.
26. The output is a cleaner image where rain-induced noise is suppressed and lane/road-edge boundaries remain sharp, supporting accurate and safe real-time navigation decisions.